Authoritarian Control through the Judiciary*

Takanao Tanaka

UC Berkeley

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Authoritarian governments may use courts to punish groups perceived as political threats. We examine this using 6.4 million Chinese criminal verdicts. Ethnic minorities receive 6.8% longer sentences and face 6.5% and 14.0% higher probabilities of life imprisonment and the death penalty, respectively. These additional penalties are particularly severe for crimes associated with social unrest, for less assimilated ethnic groups, and in regions experiencing heightened social tensions. Penalties against minorities intensified in minority-dominant prefectures during Uyghur unrest and when local political leaders sought promotions. The disparities, however, declined over time and almost disappeared in 2021; we discuss the potential causes.

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1 Introduction

All authoritarian regimes must maintain social order to ensure their survival. To prevent political opponents from challenging their authority, these regimes employ a variety of tactics—ranging from conventional methods based on violence and ideological indoctrination to surveillance, to information manipulation, to censorship, to local elections, and to economic policies (Andirin et al., 2022).[1] This paper provides empirical evidence on political control through the judiciary—a function of law in authoritarian regimes that has long been discussed but has received limited empirical examination (Solomon, 1996; Pereira, 2005). We argue that authoritarian governments can use criminal law to imprison individuals or groups perceived as potential sources of unrest and, in more extreme cases, impose life imprisonment or the death penalty.

Our empirical analysis examines the treatment of ethnic-minority defendants in China's criminal justice system. Han Chinese account for approximately 91% of the population, while China officially recognizes 55 ethnic-minority groups with diverse languages, religions, customs, and cultural traditions. Because the Chinese Communist Party (CCP) faces little organized political opposition, ethnic unrest can pose a particularly salient threat to state control. The Party controls the judiciary; about 85% of judges and virtually all court presidents are Party members. This structure may lead to unfavorable treatment of ethnic-minority defendants perceived as threats to the central government.

This paper assembles, to our knowledge, the most comprehensive and detailed dataset on China's criminal justice system. Our baseline sentence-length analysis uses 6.4 million observations from publicly available criminal court records covering 2013 to 2021. Because the Supreme People's Court required local courts, in principle, to publish all criminal verdicts, this dataset covers nearly the universe of criminal sentencing in China. Using text-processing and machine-learning methods, we extract an extensive set of detailed case and defendant characteristics from unstructured, nonstandardized judicial documents. Specifically, our data distinguish unusually narrow crime categories and capture hundreds of legally relevant aggravating and mitigating factors—detail rarely available in sentencing studies.

Drawing on these comprehensive data and comparing observably similar Han and ethnic-minority defendants, we find that minority defendants receive 6.8% longer sentences and are 6.5% and 14.0% more likely to receive life and death sentences, respectively. For comparison, the estimated disparities are of the same order of magnitude as the Black–White sentencing

[1]For reviews, see Gehlbach et al. (2016), Guriev and Treisman (2019), and Egorov and Sonin (2024). For specific tactics, see Olson (1993) on violence; Adena et al. (2015) and Yanagizawa-Drott (2014) on ideological indoctrination; Xu (2021) and Beraja et al. (2023) on surveillance; Rozenas and Stukal (2019) on information manipulation; King et al. (2013) and Chen and Yang (2019) on censorship; Martinez-Bravo et al. (2022) on local elections; and Wen (2024) on economic policies.

disparities documented in the United States (Anwar et al., 2012; Rehavi and Starr, 2014; Cohen and Yang, 2019; Feigenberg and Miller, 2021).

The results reveal substantial heterogeneity. The minority penalty is larger for drug, violent, and sex offenses, which are often viewed as particularly serious or associated with organized criminal activities. It is also larger for minority groups associated with Islam and Christianity and for ethnic groups with a history of conflict with the Han majority—groups considered less assimilated into mainstream Chinese society. The additional penalty is also greater in poorer, high-minority-share, and high-crime prefectures, where the potential risk of social disorder may be greater. Han judges, who preside over about 90% of criminal cases, also impose larger minority penalties than minority judges.

To examine the judiciary's role in maintaining political control, we ask whether courts punish ethnic minorities more harshly when threats of unrest increase (Costa and Zhao, 2025). Uyghur separatist movements in Xinjiang provide a unique setting to study such possibilities, as they may fuel fears that other minority groups will mobilize elsewhere in China (Wen, 2024). We find that during months with Uyghur conflict events, the minority–Han sentencing gap widens significantly more in high-minority-share prefectures than in low-minority-share prefectures. Sentence length, life imprisonment, and death penalty gaps widen by about 26%, 37%, and 44%, respectively, compared with the baseline gaps. The increase in sentence length is concentrated among working-age men, whom officials may view as the main participants in violent unrest. In contrast, we find little change in sentence length among women or older men.

We further investigate how local officials' career incentives shape sentencing decisions. In China's bureaucracy, prefectural leaders have strong incentives to maintain social stability because major incidents such as protests or riots may preclude promotion and, in some cases, effectively end an official's political career. We focus on China's distinctive age-based promotion system, under which officials generally lose eligibility for promotion after age 55. We find that under promotion-motivated leaders, the minority–Han sentence-length gap widens by 8% relative to the baseline minority penalty.

Our findings indicate that China's judiciary plays an active role in maintaining state control by punishing ethnic-minority defendants more harshly than observably similar Han defendants. Unfavorable treatment of ethnic minorities may have intensified over time, as the international community has raised concerns about China's harsher policies toward minorities, especially since 2017 (OHCHR, 2022). In stark contrast, our results show that minority–Han sentencing disparities decline substantially over time, with the gap in sentence length disappearing in 2021. In the conclusion, we discuss potential causes, particularly how nationwide legal reforms to increase judicial transparency and standardization may have affected ethnic sentencing disparities.

2 Background

2.1 China's Ethnic Minority Issues

Ethnic minority issues lie at the heart of social stability in China. Because the CCP remains the sole ruling party and faces virtually no organized political opposition, it regards the governance of minority communities as central to maintaining political and social stability.

Ethnicity has long been a source of political conflict in China. China's ethnic policies draw heavily on the Soviet model, granting minorities limited autonomy while seeking to prevent territorial secession. Since the founding of the People's Republic of China, the central government has formally recognized rights to regional ethnic autonomy, although these rights have rarely been fully implemented. Religious freedom, cultural and linguistic preservation, and equal protection under the law have not always been guaranteed, contributing to disputes between the Chinese government and ethnic minorities (Zhang, 2011). These tensions have become increasingly visible, with unrest in Tibet and Xinjiang in 2008 and harsher policies toward Muslim minorities since 2017.[2] These policies raise concerns about unjust imprisonment or disproportionately harsh sentences for ethnic minorities.

At the same time, the Chinese government has implemented various preferential policies benefiting ethnic minorities (Sautman, 1998). These policies have included one-child policy exemptions, extra points on national college entrance exams, tuition waivers, stipends, job preferences, and tax benefits. Minorities also have the right, at least formally, to preserve their languages and cultures. Targeted poverty-alleviation and local economic-development programs provide further support. Together, these policies could lead one to expect favorable treatment for ethnic minority defendants in the judicial system.

China's broader approach to ethnic governance helps explain the tension between preferential and coercive policies. The state seeks to build a "unitary multiethnic nation," maintaining political integration while recognizing ethnic diversity and supporting minority development (Lai, 2009). These competing objectives can produce inconsistent policies across institutions. Whether, and under what circumstances, courts treat ethnic minority defendants more leniently or more harshly than comparable Han defendants is therefore ultimately an empirical question.

2.2 China's Criminal Justice System: State Control and Existing Evidence

China's criminal justice system can function as an instrument of state control for two reasons (see Appendix A-1 for details on criminal procedure). First, courts operate under extensive

[2]These policies include mass surveillance and large-scale detention and have prompted allegations of crimes against humanity (OHCHR, 2022; Al Jazeera, 2018).

political oversight and are expected to support Party objectives, including maintaining social order (Li, 2016; Ng and He, 2017). The Party influences judges' career advancement and discipline, and courts are explicitly expected to "pursue political effects" in adjudication. More than 85% of judges were Party members in 2016, as were virtually all court presidents and vice presidents (Zhao, 2016; Woo, 1999). Second, China follows the civil-law tradition, in which judges play a central role in determining facts, applying the law, and issuing rulings (Ng and He, 2017). Historically broad sentencing ranges and limited guidance give judges substantial discretion (Ash et al., 2026; Liu et al., 2026), creating a channel through which political pressure to maintain social order may lead to disproportionately severe sentences for ethnic minority defendants.

Despite widespread attention to China's criminal justice system, systematic empirical evidence on ethnic disparities remains limited. Existing quantitative studies focus on particular offenses or groups, and their findings are mixed. Some find more lenient treatment of ethnic minorities (Peng and Cheng, 2022; Tian, 2023), others find harsher treatment (Hou and Truex, 2022; Steinert, 2023), and others find no significant disparity (Li et al., 2018).

3 Data

3.1 A Near-Complete Database of Criminal Court Judgments

We assemble a comprehensive database of criminal sentencing judgments in China from China Judgments Online (CJO) (CJO, 2021). Since 2014, the Supreme People's Court has required Chinese courts to publish judgments online. Because courts sometimes delay publication and CJO sometimes deletes judgments, we automatically collect and archive CJO judgments daily.

We use text-processing and deep-learning methods to transform millions of unstructured and nonstandardized judgments into a structured dataset containing an exceptionally rich set of variables describing each case (Appendix B-1). The data cover 3,506 courts and 49,187 judges. We extract court and judge identifiers, sentencing dates, and defendants' age, gender, ethnicity, education, occupation, and criminal history. We also predict judges' ethnicity, classifying 9.3% as ethnic minorities (see Appendix D for details). Cases span 427 crime types, providing a much finer classification than those used in existing studies. We extract hundreds of legally relevant aggravating and mitigating factors. Crime-specific measures include the number of victims killed in homicide cases, injury severity in assault cases, drug types and quantities, and stolen-property values. Common factors include guilty pleas, voluntary surrender, forgiveness by the victim, and defendants' motives, mental disorders, and disabilities (Table B1). Because courts convict more than 99.99% of defendants, we measure sentence severity using the length of fixed-term imprisonment and indicators for life

imprisonment and the death penalty.

We construct the main analytical sample in the following way (Appendix B-2). First, CJO does not publish juvenile cases or certain politically sensitive cases. Second, we focus on first-instance criminal judgments because more than 95% of second-instance proceedings leave the original judgment unchanged. Finally, we require defendants' ethnicity to be observed and construct outcome-specific estimation samples. The pooled baseline sample contains 6,506,690 cases. It represents 74.2% of the 8,769,103 input records. The pooled baseline sample is broadly similar to the input dataset in observed defendant and case characteristics (Table B2, columns (5)–(8)), partly alleviating concerns about sample selection. See Table B2 for summary statistics.

3.2 Uyghur Unrest and Prefectural Political Leaders

We manually construct a dataset of protests and violent conflict associated with Uyghur unrest in Xinjiang using reports from Wikipedia, Human Rights Watch, and BBC News (Wikipedia, n.d.; HRW, n.d.; BBC News, n.d.). The dataset contains 83 events occurring between 2013 and 2017, and we plot the monthly event counts in Panel B of Figure F1. Our sources report no events after 2017, when the government substantially intensified its repression in Xinjiang.

We compile biographical data on 3,030 prefectural mayors and Party secretaries serving between 2014 and 2021, using official résumés and government appointment and dismissal reports to identify leadership transitions. Following the age-based promotion system described in Section 6.1, we classify leaders younger than 55 when taking office as "promotion-seeking" and those aged 55 or older as "less promotion-motivated."

Prefectural minority population shares are measured using the 2010 census (Panel A of Figure F1).

3.3 Unconditional Sentencing Disparities

Figure 1 presents time trends of unconditional disparities in sentence length (Panel A) and the probabilities of life imprisonment (Panel B) and the death penalty (Panel C) between Han and ethnic minority defendants, with bars showing the number of cases in each group. On average, ethnic minority defendants receive harsher penalties than Han defendants. These unconditional disparities appear to narrow rapidly over the study period.

4 Ethnic Minorities in China's Criminal Justice System

4.1 Measuring Conditional Ethnic Disparities in Criminal Sentencing

We measure ethnic disparities in sentencing following Rehavi and Starr (2014), Cohen and Yang (2019), and Feigenberg and Miller (2021). We compare cases within the same crime categories and jurisdictions and before the same judges, while controlling for detailed case circumstances:

$$Sentence_{icpjt} = \alpha + \beta Minor_i + \boldsymbol{X}'_{icpjt}\theta + \lambda_c + \gamma_p + \sigma_j + \delta_t + \epsilon_{icpjt}. \quad (1)$$

The dependent variable, $Sentence_{icpjt}$, is either sentence length in months, an indicator for life imprisonment, or an indicator for the death penalty. Defendant i's case is decided by judge j in court c, located in province p, during year-month t. Because life imprisonment and the death penalty can be imposed only for severe offenses, we estimate the corresponding specifications using the relevant subsamples. $Minor_i$ equals one for an ethnic minority defendant and zero for a Han defendant. The specification includes court fixed effects (λ_c), province fixed effects (γ_p), judge fixed effects (σ_j), and year-month fixed effects (δ_t). These fixed effects account for differences in legal practices across jurisdictions, nationwide changes in laws and legal practices over time, and time-invariant differences across judges.

The vector $\boldsymbol{X}_{icpjt}$ contains an extensive set of defendant and case characteristics described in Section 3, including indicators for 427 crime types and the aggravating and mitigating factors collected from the judgments. We also interact relevant case indicators with quantitative measures. For example, we interact an indicator for heroin possession with the weight of the drugs involved. These controls allow us to compare Han and ethnic minority defendants within narrowly defined crime types and under similar offense circumstances.

The coefficient of interest, β, captures the average difference in sentencing outcomes between ethnic minority and Han defendants, conditional on the included covariates and fixed effects. Standard errors are clustered at the court level.

We do not adopt methods based on the quasi-random assignment of cases to judges for two reasons. First, conventional judge-leniency designs often estimate the effects of judges' decisions, such as imprisonment (Bhuller et al., 2020). Our interest, however, is in sentencing differences by ethnicity rather than the effects of these decisions. Second, Arnold et al. (2022) assess racial discrimination in bail decisions using whether released defendants miss court or are rearrested.[3] However, we have no comparable outcome measure for sentencing, as whether a sentence is appropriate is essentially a normative question.

[3]The model assumes judges seek to prevent pretrial misconduct, which is observed only among released defendants.

4.2 Five Facts about Ethnic Minorities in China's Judiciary

4.2.1 Fact 1: Harsher penalties for ethnic minorities

Ethnic minority defendants receive significantly harsher penalties than observably similar Han defendants. Figure 2 shows that minority defendants receive 6.8% (1.15 months) longer sentences than observably similar Han defendants and are 6.5% (0.084 pp) and 14.0% (0.057 pp) more likely to receive life imprisonment and the death penalty, respectively (1. Baseline).[4] These differences are economically meaningful. Eliminating the disparity would have reduced aggregate prison terms for ethnic minority defendants by approximately 77,499 years and the numbers of life and death sentences by 256 and 197, respectively, during our study period.[5] For comparison, U.S. studies find that Black defendants receive sentences 10–20% longer than non-Black defendants (Anwar et al., 2012; Rehavi and Starr, 2014; Cohen and Yang, 2019; Feigenberg and Miller, 2021). Panel A of Table E1 reports the estimates.

In China, differences in crime and case characteristics explain a large share of the unconditional sentencing disparity between Han and ethnic minority defendants. In Table E2, we sequentially add controls across columns (1)–(6). Without controls, minority defendants receive sentences that are 2.90 months (17.2%) longer than those of Han defendants (column (1)). Adding crime-type and case controls reduces the unconditional sentence-length gap by about 75%, to 0.71 months (column (2)). The remaining controls, including province, court, and judge fixed effects, produce comparatively smaller changes. This pattern is consistent with U.S. evidence that offense and case characteristics explain much of the unconditional Black–White sentencing gap (Rehavi and Starr, 2014). For life imprisonment and the death penalty, however, province fixed effects play a larger role in reducing the disparities (Panels B and C).

4.2.2 Fact 2: Larger minority penalties for severe and organized crimes

The "minority penalty"—the additional penalty imposed on ethnic minority defendants relative to observably similar Han defendants—is more pronounced for violent crimes (2 in Figure 2), drug-related offenses (3), and sex crimes (4), all of which the literature classifies as severe offenses (Schanzenbach, 2015). We obtain these results by estimating the model separately for each crime category. Panel B of Table E1 reports the estimates.

We find particularly large minority penalties for drug crimes. In China, drug crimes

[4] Among Han defendants in the respective baseline estimation samples, mean sentence length is 16.89 months, and the rates of life imprisonment and the death penalty are 1.300% and 0.405%, respectively. The latter two samples include only offenses eligible for the corresponding penalty.

[5] We assume records with missing ethnicity have the observed minority share and excluded defendants face the estimated disparities. Totals are provisional because counts and coefficients use different data versions.

are a significant concern (Biddulph, 2002; Zhang and Chin, 2015), and ethnic minorities and minority criminal syndicates are disproportionately involved. For example, minority-dominated villages in Yunnan Province, which borders Myanmar—a country that supplies more than 90% of the heroin in Chinese markets—serve as major hubs for drug trafficking. For drug crimes, ethnic minority defendants receive 5.3% (1.60 months) longer sentences and are 11.2% (0.145 pp) and 19.2% (0.078 pp) more likely to receive life imprisonment and the death penalty, respectively (Figure 2, 3. Drug Crime).

4.2.3 Fact 3: Harsher penalties for less assimilated ethnic groups

Ethnic minority groups that are less assimilated into Han culture receive significantly harsher penalties in China's criminal justice system. Figure 2 reveals sizable additional penalties for ethnic groups that practice distinct religions. Specifically, the Chinese government views Christianity and Islam as potential political rivals because of their foreign origins, distinct cultural and linguistic traditions, and their own "spiritual leaders." Defendants from Christian ethnic groups (7) receive 9.2% (1.55 months) longer sentences than observably similar Han defendants and are 8.6% (0.111 pp) and 18.5% (0.075 pp) more likely to receive life imprisonment and the death penalty, respectively. The corresponding disparities for Muslim ethnic groups (8) are 7.4% (1.26 months), 6.2% (0.080 pp), and 16.6% (0.067 pp). By contrast, defendants from ethnic groups that practice other religions, primarily Taoism, or no religion (9) face smaller sentencing disparities. Panel C of Table E1 reports the estimates.

We also find that defendants from ethnic minority groups with a history of conflict with the Han majority receive significantly more severe penalties. Compared with observably similar Han defendants, they receive sentences that are 8.8% (1.48 months) longer and are 9.2% (0.12 pp) and 25.0% (0.10 pp) more likely to receive life imprisonment and the death penalty, respectively (Figure 2, 10).

Figure E1 further reports ethnicity-specific minority penalties among the 20 largest ethnic groups, with ethnic groups ordered by the magnitude of the estimated disparity. Although the estimates are somewhat noisy, presumably because some groups have relatively few cases, the largest penalties fall on the Dai, Tibetan, and Yi groups for sentence length (Panel A); the Dai, Hani, and Yi for life imprisonment (Panel B); and the Dai, Yi, and Yao for the death penalty (Panel C). The Dai ethnic group is concentrated in Yunnan, a major drug-trafficking hub. These patterns are consistent with larger penalties for Christian groups (Yi and Hani), Muslim groups (Hui), and groups with a history of conflict with the Han majority (Tibetans and Yi).

We find possible favoritism toward Uyghur defendants (Figure E2, Panels A–C), but this finding should be interpreted with caution. Uyghurs account for 0.81% of China's population in 2015 but only 0.08% of defendant observations in our data (Panel D). This gap raises

concerns about selective verdict publication, as politically sensitive cases may be excluded from CJO. Alternatively, it may reflect differences in how frequently Uyghur individuals appear in court and receive criminal sentences.

4.2.4 Fact 4: Larger minority penalties in areas with greater social tensions

Figure 2 shows that ethnic minority defendants face larger additional penalties in prefectures with lower incomes (13), larger ethnic minority populations (14), and higher crime rates (16).[6] These regional characteristics are closely correlated. Areas with larger ethnic minority populations are concentrated in western and southern China, where incomes are lower and crime rates are higher. These areas may also face a greater risk of social unrest because larger ethnic minority populations reduce the cost of collective action and lower incomes reduce the opportunity cost of participation (Cantoni et al., 2019). Panel D of Table E1 reports the estimates.

4.2.5 Fact 5: Han judges account for most disparities

Han judges predominantly drive the minority penalty (Figure E3). Across all three outcomes, Han judges impose substantially larger additional penalties on ethnic minority defendants than do ethnic minority judges. The estimated disparities are approximately three times as large among Han judges. For example, the estimated disparity in sentence length is 7.1% among Han judges and 2.8% among ethnic minority judges (column (2) of Table E1). Because Han judges account for 91% of judges, close to their share of the national population, they account for most of the observed minority penalty. Nevertheless, ethnic minority judges also impose harsher penalties on ethnic minority defendants, although the additional penalties are smaller. Panel E of Table E1 reports the estimates.

4.3 The Role of Prosecutors in Ethnic Sentencing Disparities

We supplement our analysis by examining prosecutors, who are also key actors in the criminal justice system, because judges alone may not account for the entire minority penalty in case outcomes (Rehavi and Starr, 2014; Tuttle, 2021; Baron et al., 2024). Prosecutors can recommend sentences to the court, and these recommendations may reflect bias in evidence collection or charging (Bohren et al., 2025). Our data report these recommendations for about 16% of cases (see Appendix B-1 for details). We use them as the dependent variables in equation (1).

[6]For each characteristic, we divide prefectures into high and low groups at the median and estimate the baseline equation separately for each group. Income and ethnic minority shares are measured using the 2010 census, and crime rates are calculated using the number of cases in our dataset.

We find that prosecutors recommend harsher sentences for ethnic minority defendants than for observably similar Han defendants, although these disparities are smaller than those in final sentences. Figure E3 shows that ethnic minority defendants receive recommended prison terms that are 3.0% longer and are 3.5% and 8.5% more likely to receive recommendations for life imprisonment and the death penalty, respectively. These findings indicate that ethnic minority defendants are treated less favorably at multiple stages of the criminal process. Panel F of Table E1 reports the estimates.

5 Uyghur Unrest, Minority Population, and Minority Penalties

5.1 A Difference-in-Differences-in-Differences Approach

This section tests whether courts impose a larger minority sentencing penalty when the perceived threat of ethnic unrest increases. Following Wen (2024), we use conflict events involving Uyghur separatists and the Chinese government in Xinjiang as shocks to the nationwide salience of ethnic unrest. We expect the perceived threat of unrest to increase during conflict periods, particularly in locations with larger minority populations.

We estimate a difference-in-differences-in-differences (DDD) specification that combines variation across defendants, locations, and time. Specifically, we compare minority and Han defendants, prefectures above and below the median minority population share, and year-months with and without a conflict event in Xinjiang:

$$\begin{aligned} Sentence_{icpjt} = \alpha &+ \beta_1 Minor_i \times Unrest_t \times MinorPop_p \\ &+ \beta_2 Minor_i + \beta_3 Minor_i \times Unrest_t \\ &+ \beta_4 Minor_i \times MinorPop_p + \beta_5 Unrest_t \times MinorPop_p \\ &+ X_{icpjt} + \lambda_c + \sigma_j + \delta_t + \epsilon_{icpjt}. \end{aligned} \tag{2}$$

Here, $Unrest_t$ is an indicator equal to one if at least one conflict event occurs in Xinjiang in year-month t. $MinorPop_p$ is an indicator equal to one if the minority population share in prefecture p exceeds the sample median. We use the total minority population share, rather than the Uyghur population share, because Uyghurs constitute less than 1% of the population outside Xinjiang, unlike in the setting examined by Wen (2024). The two-way interaction terms allow both the minority–Han sentencing gap and the association between conflict periods and sentencing outcomes to differ between high- and low-minority-share prefectures. The year-month fixed effects absorb the main effect of $Unrest_t$, while the court fixed effects absorb the main effect of $MinorPop_p$.

Our coefficient of interest, β_1, measures whether the minority–Han sentencing gap increases more during conflict periods in high-minority-share prefectures than in low-minority-share prefectures. A positive β_1 would therefore indicate that minority defendants receive

relatively harsher sentences when concerns about ethnic unrest are heightened and that this response is stronger in prefectures where ethnic issues are likely to be more salient. Identification requires that, in the absence of the Xinjiang conflict events, the difference in the minority–Han sentencing gap between these two groups of prefectures would have evolved similarly during the corresponding months. We cluster standard errors at the prefecture level.

5.2 Judicial Responses to Ongoing Unrest

We find that courts impose harsher penalties on ethnic minorities in response to heightened threats of unrest. The coefficient on the triple interaction ($Minor_i \times Unrest_t \times MinorPop_p$) is 0.301 in column (1) of Table 1. This estimate indicates that, during periods of Uyghur unrest, the minority–Han gap in sentence length increases by an additional 0.301 months in high-minority-share prefectures relative to low-minority-share prefectures. This increase is about 0.26 times the baseline penalty. The coefficients for life imprisonment and the death penalty are 0.031 (column (5)) and 0.025 (column (6)), respectively, corresponding to approximately 37% and 44% of the respective baseline disparities.

The effect on sentence length is driven primarily by working-age men—major actors in violent unrest. Following Wen (2024), we estimate the specification separately by defendants' gender and age in columns (2)–(4) of Table 1. The DDD coefficient for men aged 20–59 is 0.344 (column (2)), similar to the full-sample estimate. In contrast, the coefficients for men aged 60 or older (column (3)) and women (column (4)) are statistically indistinguishable from zero, with point estimates of 0.001 and −0.020, respectively.[7] This pattern is consistent with courts responding more strongly to defendants viewed as potential sources of unrest. Minority men account for approximately 4% of China's population but more than 45% of political prisoners in the CECC database (NBSC, 2000; CECC, 2019). Our findings suggest that China's local governments use both carrots and sticks to manage unrest. When the risk of conflict spreading rises, state firms hire more minority men (Wen, 2024), while local courts impose harsher sentences on this group.

[7] We do not conduct the corresponding subgroup analyses for life imprisonment or the death penalty because fewer than 600 older male defendants in our sample received these sentences. We also cannot examine the extensive margin of conviction because courts convict more than 99.9% of defendants.

6 Local Leaders' Career Incentives and Judicial Sentencing

6.1 A Difference-in-Differences-in-Differences Approach

Prefectural Party secretaries and mayors are evaluated on economic development, social stability, and other dimensions of local performance. Serious failures in stability maintenance can preclude promotion under the "one-vote veto" system, even when performance on other objectives is strong. Leaders with stronger promotion incentives may therefore place greater emphasis on preventing unrest and encourage courts to impose harsher penalties on ethnic minorities (de Janvry et al., 2023).

We test this hypothesis using China's age-dependent promotion system. Prefectural leaders are generally expected to "step back from frontline duties" at age 55, before formal retirement. Those who take office before this threshold generally retain opportunities for promotion, whereas those who take office afterward face sharply diminished prospects for advancement (Yao and Zhang, 2015; Yu et al., 2016; Campante et al., 2023).

We estimate DDD specifications analogous to those in Section 5.1:

$$\begin{aligned} Sentence_{icpjt} = \alpha &+ \beta_1 Minor_i \times Motivated_{pt} \times MinorPop_p \\ &+ \beta_2 Minor_i + \beta_3 Motivated_{pt} \\ &+ \beta_4 Minor_i \times Motivated_{pt} + \beta_5 Minor_i \times MinorPop_p \\ &+ \beta_6 Motivated_{pt} \times MinorPop_p + X_{icpjt} + \lambda_c + \sigma_j + \delta_t + \epsilon_{icpjt}. \end{aligned} \tag{3}$$

The variable $Motivated_{pt}$ equals one if at least one of the two principal leaders in prefecture p—the mayor or the Party secretary—was younger than 55 at the beginning of their tenure, an age associated with a substantially greater likelihood of promotion. All other variables are defined as in equation (2). The coefficient β_1 measures whether the minority–Han sentencing gap widens more under promotion-motivated leaders in prefectures with larger minority shares. Switches in prefectural leaders' promotion-incentive status are infrequent, with only 30 in total. We therefore cannot construct an adequate not-yet-treated comparison group, precluding an event-study analysis.

6.2 The Impact of Local Governments' Political Motivations on the Judiciary

We find that penalties against minority defendants become harsher when local prefectural leaders are below age 55 and may therefore have stronger incentives to seek promotion and maintain local social order. In Table 2, the coefficient of interest is 0.09 for sentence length (column (1)), corresponding to 8% of our baseline results. The coefficients for life imprisonment and the death penalty are 0.017 and 0.016, respectively, corresponding to increases of 20% and 28% relative to the respective baseline minority penalties, although both estimates are imprecise. These estimates capture sizable additional penalties associated

with stronger promotion incentives, even though leaders with limited promotion prospects remain responsible for maintaining social stability.

7 Conclusion

Using newly digitized data covering nearly the universe of published criminal verdicts in China, we provide the first comprehensive nationwide analysis of ethnic sentencing disparities. We find that ethnic minorities receive disproportionately harsher penalties, with 6.8% longer sentences and 6.5% and 14.0% higher probabilities of life imprisonment and the death penalty, respectively, compared to observably similar majority defendants. These additional penalties are particularly severe for crimes associated with social unrest, among less assimilated ethnic groups, and in regions experiencing heightened social tensions. The sentence-length penalty against minorities increases in prefectures with larger minority populations during Uyghur unrest and under local leaders with stronger promotion incentives, suggesting political roots of the disparity.

These estimates require caution for three reasons. First, our estimates alone cannot establish that discrimination drives the sentencing gap, as unobserved factors correlated with ethnicity may also explain it. Nonetheless, if courts impose harsher penalties because they associate a defendant's ethnicity with a greater risk of unrest, such decisions constitute discrimination and conflict with the principle of equal treatment under the rule of law. Second, we do not account for selection into arrest and prosecution; such data are, to our knowledge, extremely challenging to obtain in China. Third, our data omit some politically sensitive cases and therefore provide an incomplete picture of how courts treat ethnic minorities.

Our results suggest political roots of ethnic sentencing disparities. Previous studies offer two common explanations for sentencing disparities, but neither fully explains our findings. First, our results do not support in-group favoritism as the sole explanation (Shayo and Zussman, 2011; Anwar et al., 2012; Ash et al., 2021). In-group favoritism does not explain why additional penalties concentrate in crimes and ethnic groups associated with perceived threats to social order. Minority judges also impose harsher penalties on minority defendants, seemingly contradicting in-group favoritism. Second, U.S. studies suggest that local courts respond to voters' preferences (Lim, 2013; Berdejo and Yuchtman, 2013; Feigenberg and Miller, 2021). In China, however, minority penalties are larger where minorities make up a greater share of the local population, and they widen when local leaders face stronger promotion incentives. These patterns suggest a distinct political role for the Chinese judiciary: supporting the Party's goal of preventing social unrest.

Our interpretation raises a puzzle. Since 2017, international observers have raised concerns about increasingly restrictive ethnic policies in Xinjiang (OHCHR, 2022). If courts help

maintain state control, we might therefore expect ethnic sentencing disparities to widen. Our results show the opposite. Figure 3 shows that the sentence-length gap generally narrowed, with gaps in all three outcomes statistically indistinguishable from zero by 2021.

One potential explanation for the near disappearance of the ethnic sentence-length gap is China's national effort to improve the transparency and consistency of judicial decisions. These efforts include sentencing guidelines (2010, 2014, 2017), rules governing the use of guiding cases (2015), safeguards against officials' interference in individual cases (2015), online judgment-publication requirements (2016), and requirements to explain judicial reasoning (2018) (SPC, 2015b,a, 2016, 2018).

Among these measures, sentencing guidelines may have helped narrow the ethnic sentencing gap, as they directly address how judges determine sentences. The guidelines limit judges' discretion (SPC, 2014). For example, the criminal law prescribes at least seven years for trafficking either 20 or 25 grams of heroin, without specifying how much the additional five grams should increase the sentence. The guidelines specify a one-year increase in the baseline sentence (Jiangsu HPC, 2014). With these guidelines in place, judges have less discretion to impose harsher penalties on defendants based on characteristics such as ethnicity.

Evaluating the effects of these guideline reforms, however, is challenging. China introduced the guidelines on a nationwide trial basis in 2010 and formally implemented them in 2014 (Guo, 2014). We cannot credibly estimate the effects of the 2010 and 2014 reforms because mandatory online publication of judgments began only in 2014. In 2017, China revised the nationwide sentencing guidelines for 15 targeted offenses (Chen, 2017). In Appendix G, we show an estimated 0.5-month reduction in the ethnic sentencing gap for targeted offenses relative to non-targeted offenses following this nationwide reform.[8] However, drawing firm conclusions is difficult because the estimates are statistically imprecise and several legal reforms were underway.

Regardless of its causes, the remarkable decline in ethnic sentencing disparities suggests progress toward more equal treatment in China's courts. A more pessimistic possibility is that the decline reflects a shift toward other tools of state control, such as surveillance and extrajudicial detention (Xu, 2021; Beraja et al., 2023; OHCHR, 2022). Further research is needed to understand what drives this decline and whether it reflects broader progress toward equal treatment.

[8]The OLS and BJS estimates each imply a reduction equal to about 20% of the unadjusted pre-reform minority–Han gap in targeted offenses (2.93 months).

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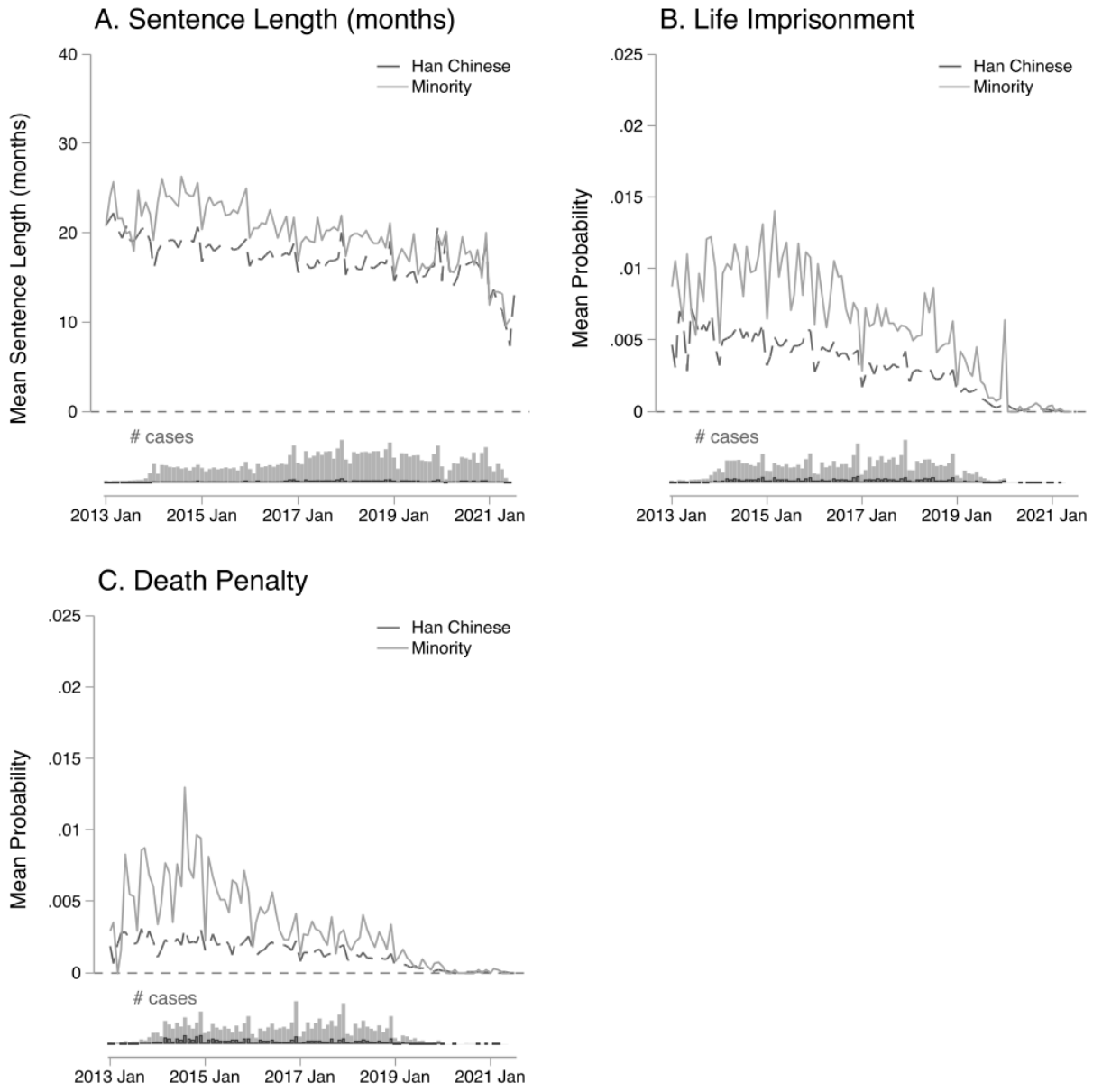
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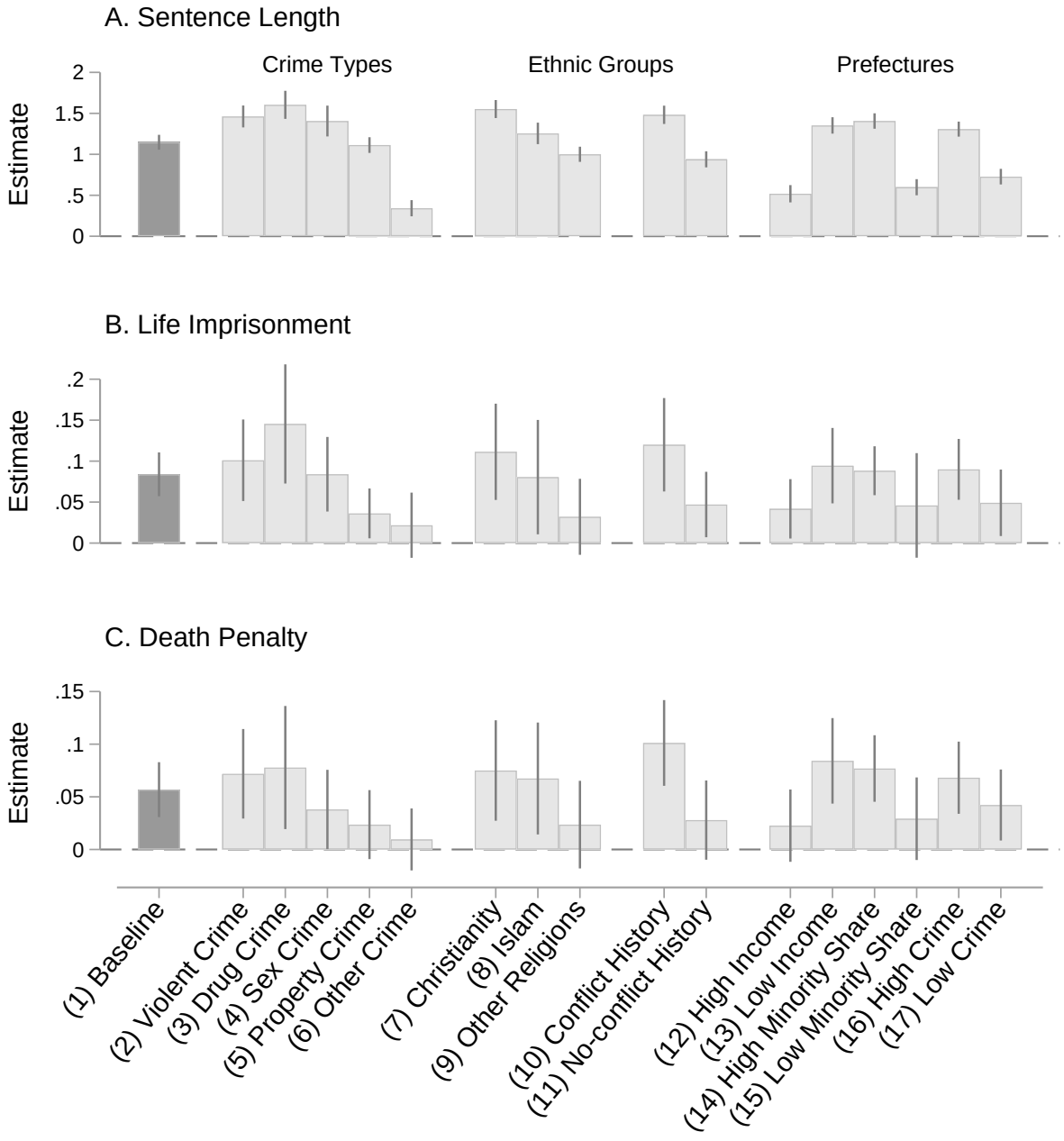
Figures

Figure 1: Trends in Average Sentencing for Han and Ethnic Minority Defendants



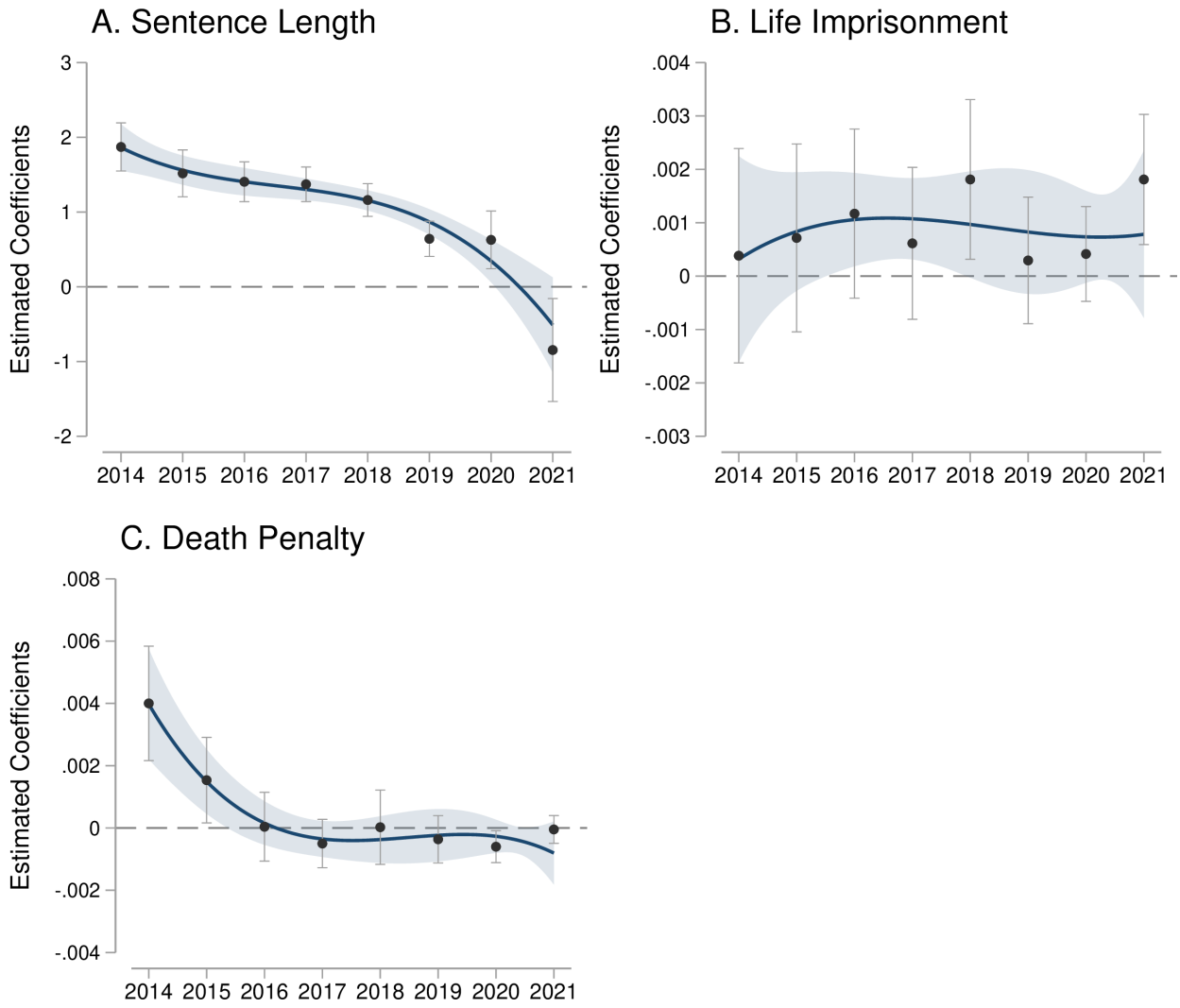
Notes: This figure illustrates the time trend of average sentencing outcomes for Han and Minority groups. Panel A displays the average imprisonment sentence length. Panel B shows the average probability of life imprisonment. Panel C presents the average probability of receiving a death sentence. The bar below represents the number of corresponding cases, with darker gray indicating Minority cases and lighter gray indicating Han majority cases.

Figure 2: Facts about Ethnic Minority Issues in China's Criminal Justice System



Notes: This figure presents key findings on the facts about ethnic minority issues in China's criminal justice system. The outcome variables are sentence length in months (Panel A), life imprisonment (Panel B), and the death penalty (Panel C). For life imprisonment and the death penalty, we restrict the samples to crime types that are eligible for these sentencing penalties. The bars denote the estimated coefficients, while the lines represent the 95% confidence intervals. The category labels below Panel C apply to all three panels. We cluster standard errors at the court level. See the main text for details. Table E1 reports the estimates and standard errors. Sentence-length coefficients are in months; life-imprisonment and death-penalty coefficients are in percentage points.

Figure 3: Minority Penalty over Time



Notes: This figure reports the estimated minority–Han sentencing gap over 2013–2021. Panel A reports sentence length in months, Panel B reports the probability of life imprisonment, and Panel C reports the probability of the death penalty. The points show coefficients from regressions that interact minority status with year indicators; whiskers show their 95% confidence intervals. The colored lines show cubic time trends estimated from individual-level regressions, and the shaded areas show their pointwise 95% confidence intervals. The years 2013 and 2014 are pooled into one initial period in both specifications. All specifications include the replication controls and fixed effects, including actual year-month fixed effects. The life-imprisonment and death-penalty samples are restricted to eligible offenses. The 2021 estimates use the available partial-year observations. Standard errors are clustered at the court level.

Tables

Table 1: Uyghur Unrest and Penalties against Ethnic Minorities

	Sentence length (months)					
	All	Men age 20–59	Men age 60+	Women	Life imprison-ment	Death penalty
	(1)	(2)	(3)	(4)	(5)	(6)
Minority × Unrest × Minority Pref	0.301***	0.344***	0.001	-0.020	0.031**	0.025*
	(0.048)	(0.076)	(0.074)	(0.063)	(0.015)	(0.014)
Observations	3,131,373	1,976,305	55,818	261,122	877,448	1,087,269
Crime category FE	Y	Y	Y	Y	Y	Y
Case characteristics	Y	Y	Y	Y	Y	Y
Defendant characteristics	Y	Y	Y	Y	Y	Y
Year-month FE	Y	Y	Y	Y	Y	Y
Province FE	Y	Y	Y	Y	Y	Y
Court FE	Y	Y	Y	Y	Y	Y
Judge FE	Y	Y	Y	Y	Y	Y

Notes: This table reports the DDD estimates from equation (2). Columns (1)–(4) report sentence-length estimates in months for all defendants, men aged 20–59, men aged 60 or older, and women, respectively. Columns (5) and (6) include all demographic groups and restrict the sample to crimes eligible for life imprisonment and the death penalty, respectively. Minority identifies ethnic-minority defendants; Unrest identifies months with a Uyghur conflict event; Minority Pref identifies prefectures at or above the configured minority-population-share cutoff. All specifications use cases from 2013–2017 and exclude Xinjiang. The relevant lower-order interactions enter every specification. Standard errors, in parentheses, are clustered by prefecture. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 2: Local Governments, Judiciary, and Penalties against Minorities

	Sentence Length	Life Imprisonment	Death Penalty
	(1)	(2)	(3)
Minority × Below Age 55 × Minority Pref	0.091***	0.017	0.016
	(0.032)	(0.075)	(0.069)
Crime category FE	Y	Y	Y
Case characteristics	Y	Y	Y
Defendant characteristics	Y	Y	Y
Year-month FE	Y	Y	Y
Province FE	Y	Y	Y
Court FE	Y	Y	Y
Judge FE	Y	Y	Y

Notes: This table presents the DDD results from equation (3). "Minority" indicates whether the defendant belongs to an ethnic minority group. "Below Age 55" equals one if at least one of the two prefectural leaders—the mayor or Party secretary—was younger than 55 at the start of their tenure. "Minority Pref" equals one if the prefectural minority population share is above the median. See the main text for further details. Standard errors, clustered at the prefecture level, are reported below the coefficients. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

ONLINE APPENDIX

Authoritarian Control through the Judiciary

Takanao Tanaka

Tanaka: Department of Agricultural and Resource Economics, UC Berkeley
(takanao_tanaka@berkeley.edu)

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A Additional Background

A-1 Criminal Procedure

China's criminal justice system assigns distinct but coordinated roles to public security organs, People's Procuratorates, and People's Courts. Police investigate suspected offenses, gather evidence, and transfer sufficiently supported cases to the procuratorate. Prosecutors review the evidence and legality of the investigation, then initiate prosecution, decline to prosecute, or request supplementary investigation. Cases prosecuted proceed to court for adjudication. Our analysis examines ethnic disparities arising after individuals enter this system.

Criminal cases are adjudicated within a four-level court hierarchy, with most first-instance trials conducted by primary or intermediate people's courts. China follows the principle that the second instance is final: a first-instance judgment becomes final unless the defendant appeals or the procuratorate files a protest. When either occurs, a higher-level court reviews the case, and its judgment is generally final.

Two features of adjudication are particularly relevant to our analysis. First, 99.994% of defendants whose cases reached adjudication in 2018 were convicted. Within the court verdicts observed in our data, the principal variation therefore concerns sentence severity rather than conviction or acquittal. Second, judges determine the relevant facts and apply statutory sentencing rules while retaining considerable discretion within prescribed ranges and in applying aggravating or mitigating provisions. Drug offenses illustrate this discretion: statutory ranges depend on drug type and quantity, while the precise sentence also depends on other circumstances of the offense.

China's treatment of guilty pleas also matters for sentencing. Unlike the negotiated plea bargaining prevalent in the United States, China's system of leniency for confession and acceptance of punishment allows defendants to agree to prosecutorial sentencing recommendations while requiring court adjudication. Defendants who confess and accept punishment generally receive more lenient sentences, and use of this system has expanded in recent years.[1] We account for this variation through a corresponding control variable.

[1] See https://www.chinalawtranslate.com/en/a-plea-for-greater-reform-and-engagement/.

B Data Appendix

B-1 Variable Construction

We explain the detailed variables used in our analysis. First, we extract institutional and demographic characteristics. The judgments identify the court and judge responsible for each case, as well as the year and month of sentencing. Our data cover 3,506 courts and 49,187 judges. We also extract defendants' age, gender, ethnicity, education, occupation, and criminal history.

Second, we construct a detailed classification of criminal offenses. We classify cases into 427 distinct crime types. For example, we divide drug-related offenses into 13 categories and separately classify illegal possession, trafficking, production, cultivation, harboring drug offenders, and facilitating drug use.

Third, we extract case characteristics that criminal law recognizes as aggravating or mitigating sentencing factors. These include both crime-specific measures of severity and factors applicable across crimes. Crime-specific measures include the number of victims killed in homicide cases, the severity of injuries in assault cases, the types and quantities of drugs involved in drug-related offenses, and the value of stolen property in theft cases. Common factors include guilty pleas, voluntary surrender, forgiveness by the victim, and information on defendants' attitudes, motives, mental disorders, and disabilities. Table B1 provides examples of these controls.

Finally, we extract judicial outcomes and prosecutors' sentencing recommendations. Approximately 20% of crime types permit life imprisonment or the death penalty. We also observe prosecutors' recommended sentences in approximately 15.7% of cases, allowing us to examine whether ethnic disparities arise in prosecutorial recommendations or judicial sentencing decisions. Under the 2010 joint rules, recommendations are reported when the facts are clear, the evidence is credible and sufficient, and relevant sentencing factors have been considered (SPC et al., 2020).

B-2 Construction of Main Sample

We describe how we construct the sample. Since 2014, the Supreme People's Court of China has required courts at all levels to publish both newly issued and historical judgments through the platform, giving CJO broad coverage of criminal court decisions. The platform nevertheless removes large batches of judgments without public notice, including a major batch deletion in 2021. To limit the resulting data loss, we use an automated daily collection process that archives judgments before they are removed.

CJO does not provide complete coverage of Chinese court cases. The platform does not publish juvenile cases or certain politically sensitive cases. Although the Supreme People's

Court has required courts to publish judgments online since 2014, coverage varies across courts and years. Some courts also publish judgments issued before 2014, while others have incomplete coverage even after the requirement took effect. Our data contain fewer judgments from 2014–2016 than from 2017–2018. Despite these limitations, the raw CJO data contain 11.78 million judgments from 2014 to 2021, equivalent to approximately 89% of the cases reported by the Supreme People's Court.

We restrict our analysis to first-instance judgments, which account for 71% (8.4 million) of the raw data. The remaining records consist of sentence-commutation decisions (21%), second-instance judgments (6.1%), and modifications (1.9%). Excluding second-instance judgments entails little loss of independent information because more than 95% of these proceedings leave the original judgment unchanged.

The input dataset contains 8,769,103 records, of which 6,583,546 have known defendants' ethnicity. The pooled baseline sample is the union of the sentence-length, life-imprisonment, and death-penalty estimation samples, counting each case once. It contains 6,506,690 cases, or 74.2% of the input dataset. The remaining input records comprise 2,185,557 with unknown ethnicity and 76,856 with known ethnicity that are not retained in any of the three estimation samples. For the sentence-length regression, excluding 142,585 known-ethnicity records with missing sentence length and 6,816 singleton observations yields 6,434,145 cases. The life-imprisonment and death-penalty regressions use 1,446,093 and 1,957,249 cases, respectively, restricted to eligible offenses. Life and death sentences are retained in the pooled sample even though their sentence length is missing.

Table B2 combines summary statistics and the sample comparison. Columns (1)–(4) split the pooled baseline sample into 5,875,662 Han and 631,028 ethnic minority defendants. Columns (5)–(6) describe their combined sample of 6,506,690 cases, and columns (7)–(8) describe all 8,769,103 input records, including those with unknown ethnicity. All three pooled-sample groups include life and death sentences. Statistics use nonmissing observations for each variable. The baseline and full input samples are broadly similar in observed defendant and case characteristics, partly alleviating concerns about selection due to missing-data exclusions.

Table B1: Summary of Mitigating and Aggravating Factors

Crime Type	Variables	Sources
A. General Factors		
Variables Mentioned in Criminal Law	Committing a crime due to negligence, Mental illness, Deaf/dumb/blind, Preparation for the crime, Attempted crime, Discontinued crime, Leading role in the crime, Secondary role in the crime, Coerced into committing the crime, Abetting others to commit crimes, Social impact of the crime, Repeat offenses, Voluntarily surrendering, Providing useful information in solving the crime	CLPRC (2011)
Variables Not Mentioned in Criminal Law	Voluntary confession, Good attitude in confession, Regret for committing the crime, Compensation to the victim, Forgiveness from the victim, Good behavior in custody, Criminal settlement agreement, Targeting vulnerable people, Committing crimes during disasters, Large subjective malignancy, Small subjective malignancy, Bad motivation for committing the crime, Reasonable motivation for the crime, Using cruel methods, Crime committed in public, Severe consequences	GOSPPC (2017), IBCP (2010)
B. Crime-specific Factors		
Theft	Value of stolen items, Stealing from a home, Pickpocketing, Repeated theft, Stealing with weapons	CLPRC (2011)
Drug Trafficking	Type of drug, Weight of the drug, Being the key player, Armed trafficking, Resisting arrest with violence, International drug trafficking, Selling drugs to minors, Exploiting minors in trafficking, Recommitting drug-related crimes, Hired to transport drugs, Induced to sell higher quantities by authorities, Low drug content	CLPRC (2011)
Assault	Number of slightly injured people, Number of minor injuries (Class 1 & 2), Number of serious injuries (Class 1 & 2), Number of deaths	CLPRC (2011)
Homicide	Number of deaths	CLPRC (2011)

Notes: Data Source CLPRC (2011): Criminal Law of the People's Republic of China (2011 Amendment); GOSPPC(2017): Guiding Opinions on the Sentencing Procedures of the People's Courts (2017 Version); IBCP(2010): Implementing a balanced criminal policy of severity and leniency in cases of intentional homicide, injury, and organized crime with mafia-like characteristics (《在审理故意杀人、伤害及黑社会性质组织犯罪案件中切实贯彻宽严相济刑事政策》, 2010)

Table B2: Summary Statistics and Sample Comparison

	Han Chinese (N = 5,875,662)		Ethnic minority (N = 631,028)		Baseline sample (N = 6,506,690)		All (N = 8,769,103)	
	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. Defendant characteristics								
Age	37.9	(11.2)	34.9	(10.7)	37.6	(11.2)	37.6	(11.2)
Gender (male = 1)	0.908	(0.290)	0.919	(0.272)	0.909	(0.288)	0.909	(0.288)
Primary school and below	0.277	(0.448)	0.424	(0.494)	0.292	(0.454)	0.290	(0.454)
Secondary school	0.490	(0.500)	0.416	(0.493)	0.482	(0.500)	0.481	(0.500)
High school / junior college	0.203	(0.402)	0.137	(0.344)	0.196	(0.397)	0.198	(0.398)
Above 4-year college	0.030	(0.172)	0.023	(0.151)	0.030	(0.170)	0.031	(0.173)
High-paying job (yes = 1)	0.101	(0.301)	0.061	(0.239)	0.097	(0.296)	0.090	(0.287)
B. Case characteristics								
Offense level (1–3)	1.797	(0.565)	1.799	(0.507)	1.797	(0.559)	1.798	(0.546)
Offense level 1	0.282	(0.450)	0.250	(0.433)	0.278	(0.448)	0.271	(0.444)
Offense level 2	0.639	(0.480)	0.702	(0.457)	0.646	(0.478)	0.661	(0.473)
Offense level 3	0.079	(0.270)	0.048	(0.214)	0.076	(0.264)	0.069	(0.253)
Criminal history (yes = 1)	0.168	(0.374)	0.173	(0.378)	0.168	(0.374)	0.166	(0.372)
Guilty plea (yes = 1)	0.087	(0.281)	0.054	(0.226)	0.083	(0.276)	0.074	(0.262)
Surrender (yes = 1)	0.466	(0.499)	0.429	(0.495)	0.462	(0.499)	0.465	(0.499)
Confession (yes = 1)	0.699	(0.459)	0.695	(0.460)	0.698	(0.459)	0.696	(0.460)
Plaintiff forgiveness (yes = 1)	0.221	(0.415)	0.183	(0.387)	0.217	(0.412)	0.214	(0.410)
C. Case resolution								
Sentence length (months)	16.9	(25.2)	19.8	(29.5)	17.2	(25.7)	17.2	(25.7)
Life imprisonment (yes = 1)	0.0027	(0.0517)	0.0057	(0.0751)	0.0030	(0.0544)	0.0031	(0.0558)
Death penalty (yes = 1)	0.0012	(0.0341)	0.0029	(0.0535)	0.0013	(0.0364)	0.0016	(0.0404)

Notes: Columns (1)–(4) split the pooled baseline sample into Han and ethnic minority defendants. Columns (5)–(6) pool observations used in at least one baseline sentence-length, life-imprisonment, or death-penalty regression, counting each case once. Columns (7)–(8) describe all supplied records, including those with unknown ethnicity. Header counts are sample totals; means and standard deviations use nonmissing observed values for each variable. Sentence-length statistics use records with a term reported in months; life-imprisonment and death-penalty indicators use all records in each displayed sample. See Appendix B-2 for sample construction.

C Verdict Example

The example below illustrates the information we extract from criminal judgments. The first section records defendant characteristics, the second describes the offense, and the third reports sentencing factors and penalties. We provide an English translation followed by the Chinese original.

C-1 English Translation

Part I

People's Court of Hepu County, Guangxi Zhuang Autonomous Region
Criminal Judgment

Public prosecution agency: Hepu County People's Procuratorate, Guangxi Zhuang Autonomous Region.
The defendant, Zhu Youchuan, is a Han Chinese man born on July 12, 1984, in Hepu County, Guangxi. He has a junior high school education. He was unemployed and lived in Hepu County, Guangxi. On January 17, 2018, the Yinhai Branch of the Beihai Municipal Public Security Bureau placed him in administrative detention for 12 days for drug use. On January 29, 2018, the same branch ordered him to undergo two years of compulsory drug rehabilitation in isolation. On suspicion of drug trafficking, he was placed in criminal detention by the same branch on March 7, 2018, and formally arrested on March 19, 2018. He is currently held at the Beihai No. 2 Detention Center.

Part II

The Hepu County People's Procuratorate charged the defendant, Zhu Youchuan, with drug trafficking under indictment No. 236 [2018] and brought the prosecution before this court on May 24, 2018. The court applied the simplified procedure in accordance with the law and conducted a public trial before a single judge. The Hepu County People's Procuratorate assigned prosecutor Pang Bin to appear in court to support the prosecution. The defendant, Zhu Youchuan, also appeared in court. The proceedings have concluded.

The court found that at approximately 14:00 on January 16, 2018, Chen, a drug purchaser, called a person nicknamed "Er Ge" to purchase 200 yuan worth of heroin. They agreed to make the exchange at the entrance of Hepu County People's Hospital. "Er Ge" said that he would have a person nicknamed "Twenty-six" bring the heroin to the meeting place. At approximately 14:30 that day, the defendant, Zhu Youchuan (nicknamed "Twenty-six"), arrived at the hospital entrance with heroin and called Chen to arrange the exchange at the intersection of Dinghai Road and Pedestrian Street in Lianzhou Town, Hepu County, Guangxi. After Zhu Youchuan and Chen completed the transaction there, officers from the Second Brigade of the Beihai Municipal Public Security Bureau's Anti-Drug Detachment and the Yinhai Branch's Anti-Drug Brigade apprehended them at the scene. The officers seized from Chen two packets of suspected heroin sold to him by Zhu Youchuan, with a net weight of 0.51 grams. They also seized 200 yuan in drug proceeds and six packets of suspected heroin from Zhu Youchuan; the recorded net weights were 0.38 grams, 0.42 grams, and 0.08 grams. They separately seized a mobile phone used in the offense. Testing confirmed that the seized substances contained heroin.

Part III

The defendant, Zhu Youchuan, raised no objection to these facts at trial. The facts were also supported by his household registration certificate, documentation of his apprehension, call records, evidence-collection records, the inventory of seized items, weighing records, Chen's witness testimony, Zhu Youchuan's statement, the physical and chemical examination report and notice of expert findings, records and photographs of the scene inspection, identification records, and other evidence. This evidence was sufficient to establish the facts.

This court holds that the defendant, Zhu Youchuan, violated the state's drug-control regulations by illegally selling heroin to others. His conduct constituted the offense of drug trafficking, and the prosecution's charge is established. Zhu Youchuan played the leading role in the joint drug-trafficking offense and was a principal offender. He should therefore be punished in accordance with the law for all offenses in which he participated. Because he truthfully confessed to his offenses after being apprehended, he may be given a lighter sentence in accordance with the law. The mobile phone used for drug trafficking shall be confiscated. Considering the facts, nature, circumstances, and harm to society of the defendant's offense, and pursuant to Article 347, paragraphs 1 and 4; Article 25, paragraph 1; Article 26, paragraphs 1 and 4; Article 52; Article 53; Article 67, paragraph 3; and Article 64 of the Criminal Law of the People's Republic of China, the court rules as follows:

1. The defendant, Zhu Youchuan, is sentenced to 10 months' imprisonment and fined RMB 2,000 for drug trafficking. (The term of imprisonment shall run from the date the judgment is executed. Each day of detention before execution of the judgment shall count as one day served. The term therefore runs from March 7, 2018, to January 6, 2019. The fine shall be paid within 10 days of the judgment taking legal effect. Overdue payment shall be subject to compulsory enforcement.)

2. One mobile phone transferred with the case will be confiscated and turned over to the state treasury.

If you disagree with this judgment, you may appeal to the Beihai Intermediate People's Court of Guangxi Zhuang Autonomous Region, either through this court or directly, within 10 days beginning on the day after receipt of the judgment. A written appeal must be submitted as one original and two copies.

Judge Liu Weipei
June 6, 2018
Judicial Assistant Chen Haijuan
Clerk Huang Chuntian

C-2 Chinese Original

Part I

广西壮族自治区合浦县人民法院
刑事判决书

（2018）桂0521刑初277号

公诉机关广西壮族自治区合浦县人民检察院。

被告人朱有传，男，1984年7月12日出生于广西合浦县，汉族，初中文化，无业，住广西合浦县。因吸毒于2018年1月17日被北海市公安局银海分局处以行政拘留12日；于2018年1月29日被北海市公安局银海分局决定强制隔离戒毒二年。因涉嫌犯贩卖毒品罪于2018年3月7日被北海市公安局银海分局刑事拘留，同年3月19日被逮捕。现羁押于北海市第二看守所。

Part II

合浦县人民检察院以合检公刑诉［2018］236号起诉书指控被告人朱有传犯贩卖毒品罪，于2018年5月24日向本院提起公诉。本院依法适用简易程序，实行独任审判，公开开庭审理了本案。合浦县人民检察院指派检察员庞滨出庭支持公诉，被告人朱有传到庭参加诉讼。现已审理终结。

经审理查明：2018年1月16日14时左右，购毒人陈某电话联系绰号叫“二哥”的人要购买200元毒品海洛因，并约定在合浦县人民医院门口交易，“二哥”说其叫一名绰号“二十六”的人拿毒品海洛因到交易地点。当日14时30分左右，被告人朱有传(绰号二十六)持毒品海洛因到达合浦县人民医院门口，并打电话给陈某，约定在广西合浦县廉州镇定海路与步行街交汇处进行毒品交易，当被告人朱有传和陈某在上述地点交易完成后，被北海市公安局禁毒支队第二大队与北海市公安局银海分局禁毒大队民警现场抓获，当场从陈某身上缴获被告人朱有传贩卖给其的疑似毒品海洛因2小包，经称重，净重0.51克，并从被告人朱有传身上缴获毒资人民币200元和6小包疑似毒品海洛因，经称重，净重0.38克，0.42克，0.08克，作案手机一台。经鉴定，缴获的疑似毒品检出海洛因。

Part III

上述事实，被告人朱有传在开庭审理过程中也无异议，并有户籍证明，归案证明，通话记录，提取笔录，扣押物品清单，称量笔录，证人陈某的证言，被告人朱有传的供述，理化检验报告及鉴定意见通知书，现场勘验笔录及照片，辩认笔录等证据证实，足以认定。

本院认为，被告人朱有传违反国家对毒品的管理规定，非法向他人销售毒品海洛因，其行为构成贩卖毒品罪，公诉机关指控被告人朱有传犯贩卖毒品罪罪名成立。在共同贩卖毒品犯罪中，被告人朱有传起主要作用，是主犯，依法应按其所参与的全部犯罪处罚。被告人朱有传归案后如实供述自己的犯罪事实，依法可以从轻处罚。贩卖毒品使用的手机依法应当予以没收。根据被告人犯罪的事实、性质、情节和对社会的危害程度，依照《中华人民共和国刑法》第三百四十七条第一款和第四款、第二十五条第一款、第二十六条第一款和第四款、第五十二条、第五十三条、第六十七条第三款、第六十四条之规定，判决如下：

一、被告人朱有传犯贩卖毒品罪，判处有期徒刑十个月，并处罚金人民币2000元。

（刑期从判决执行之日起计算。判决执行以前先行羁押的，羁押一日折抵刑期一日，即自2018年3月7日起至2019年1月6日止；罚金限于本判决发生法律效力之日起十日内缴纳，逾期则强制缴纳）。

二、随案移送的手机一台，予以没收，上缴国库。

如不服本判决，可在接到判决书的第二日起十日内，通过本院或者直接向广西壮族自治区北海市中级人民法院提出上诉。书面上诉的，应提交上诉状正本一份，副本二份。

审 判 员　刘伟培

二〇一八年六月六日

法官助理　陈海娟

书 记 员　黄春添

D Judge Ethnicity Prediction

The judgments report judges' names but not their ethnicities. We train a model on census records containing names and ethnicities, then use it to predict whether judges belong to the Han majority or an ethnic minority.

D-1 Building the Classification Model

The 2005 census contains names and recorded ethnicities for nearly 13 million people. We set aside people whose surnames are predominantly Han, leaving about 2.5 million people whose surnames are shared by Han and minority groups. We randomly divide this sample into 90% for training and 10% for testing.

The model uses a long short-term memory (LSTM) neural network (Hochreiter and Schmidhuber, 1997). It reads each name character by character, learns patterns associated with ethnicity, and predicts the probability that the person belongs to an ethnic minority.[2]

Minorities make up only 7.2% of the training sample. We sample minority observations more often during training to help the model learn their names. We also use focal loss, which gives more weight to names that are difficult to classify (Lin et al., 2017).

The model's reported accuracy is 75% for the sample with shared surnames. Because Han names dominate this sample, we also assess how well the model distinguishes the two groups. Figure D1 shows the receiver operating characteristic (ROC) curve: the trade-off between correctly identifying minorities and incorrectly labeling Han individuals as minorities at different classification thresholds. The area under this curve (AUC) is 0.77 in the test set; 0.5 indicates random guessing and 1 indicates perfect classification.

D-2 Predicting Judges' Ethnicities

We apply the classification procedure to judges' names and calculate the predicted Han share in each province. Figure D2 compares this share with the Han population share in the 2005 census. Provinces with larger Han populations also have larger predicted shares of Han judges ($R^2 = 0.898$).

The predictions reproduce broad geographic patterns, but this does not verify individual judges' ethnicities. Their actual ethnicities are unobserved, and judges may differ in ethnic composition from the local population.

[2]The network uses character embeddings, 10% dropout, an LSTM layer, two linear layers separated by a ReLU activation, and a final softmax layer. We train it for 3,500 iterations with batches of 64 names.

Figure D1: Classification Model Performance: ROC Curve

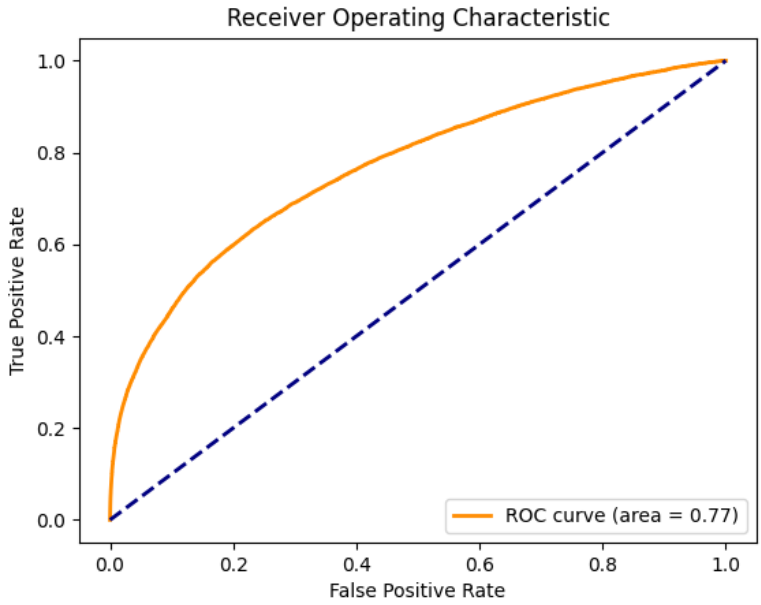
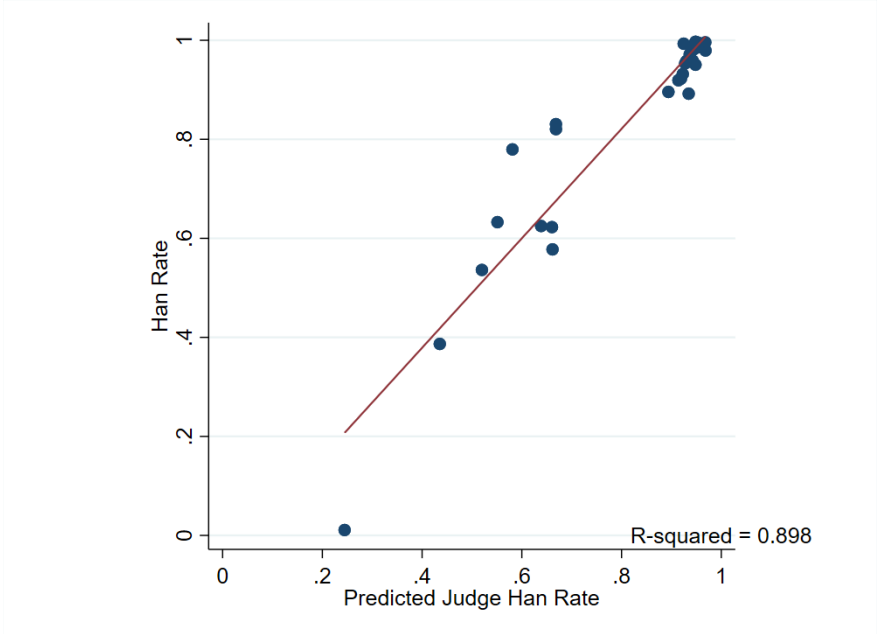
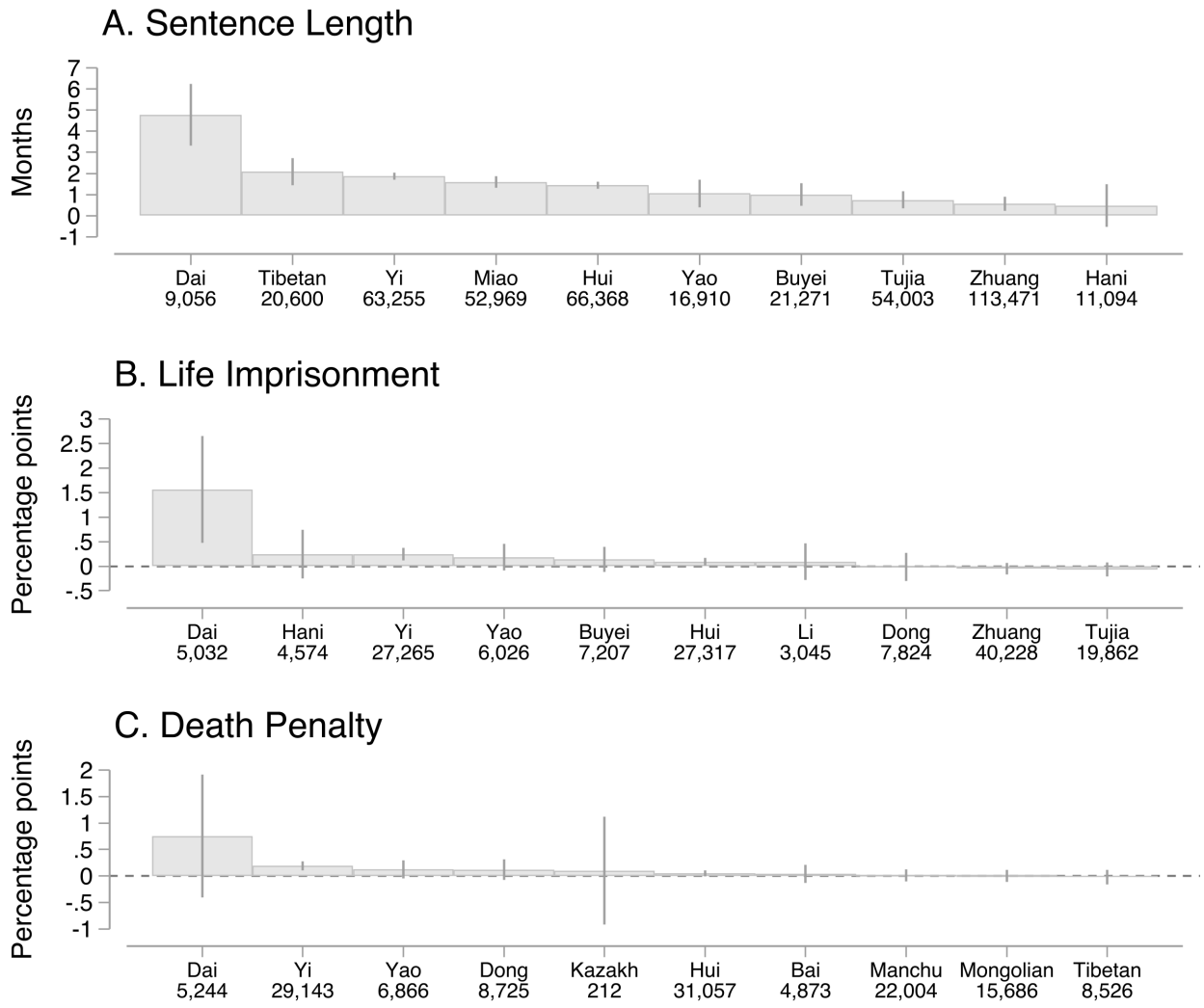


Figure D2: Predicted Han Share among Judges and Han Population Share



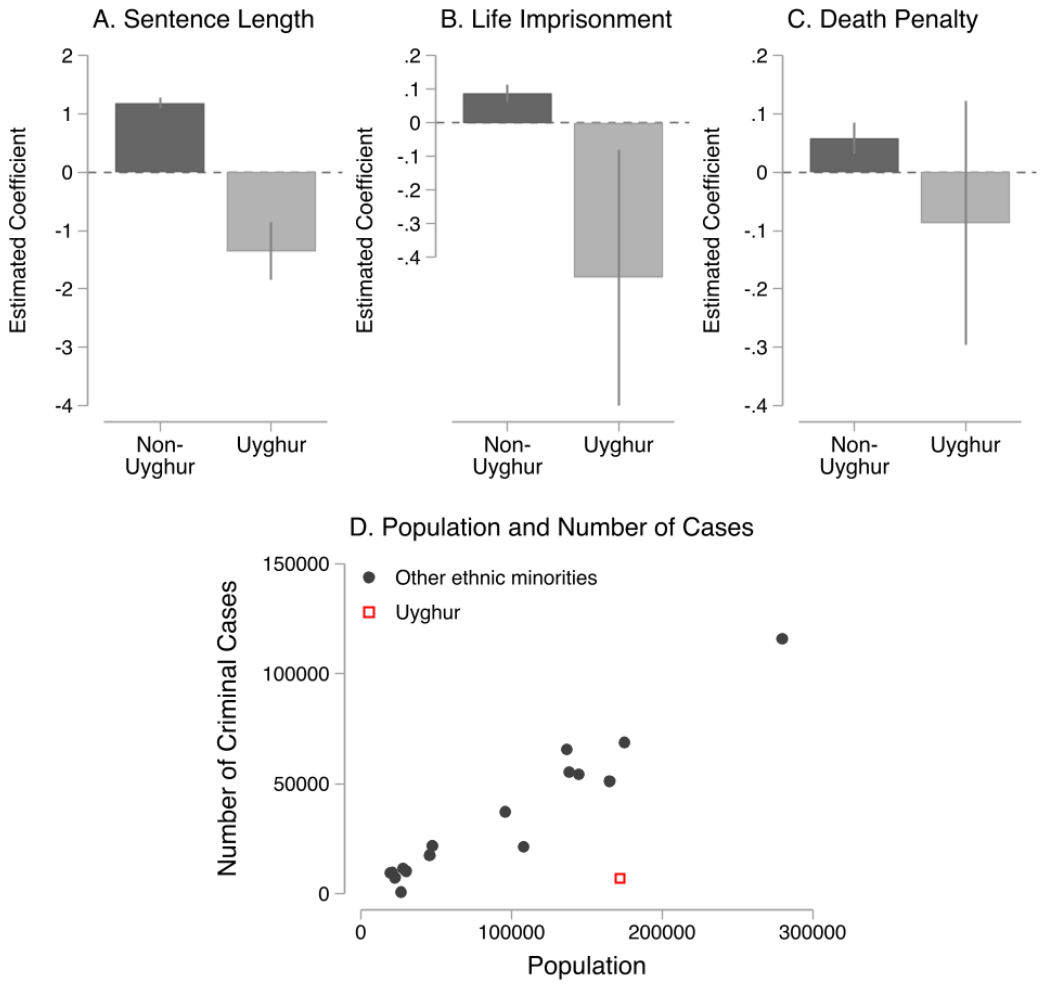
E Additional Results on Sentencing Disparities

Figure E1: Minority Penalties among the 20 Largest Ethnic Groups



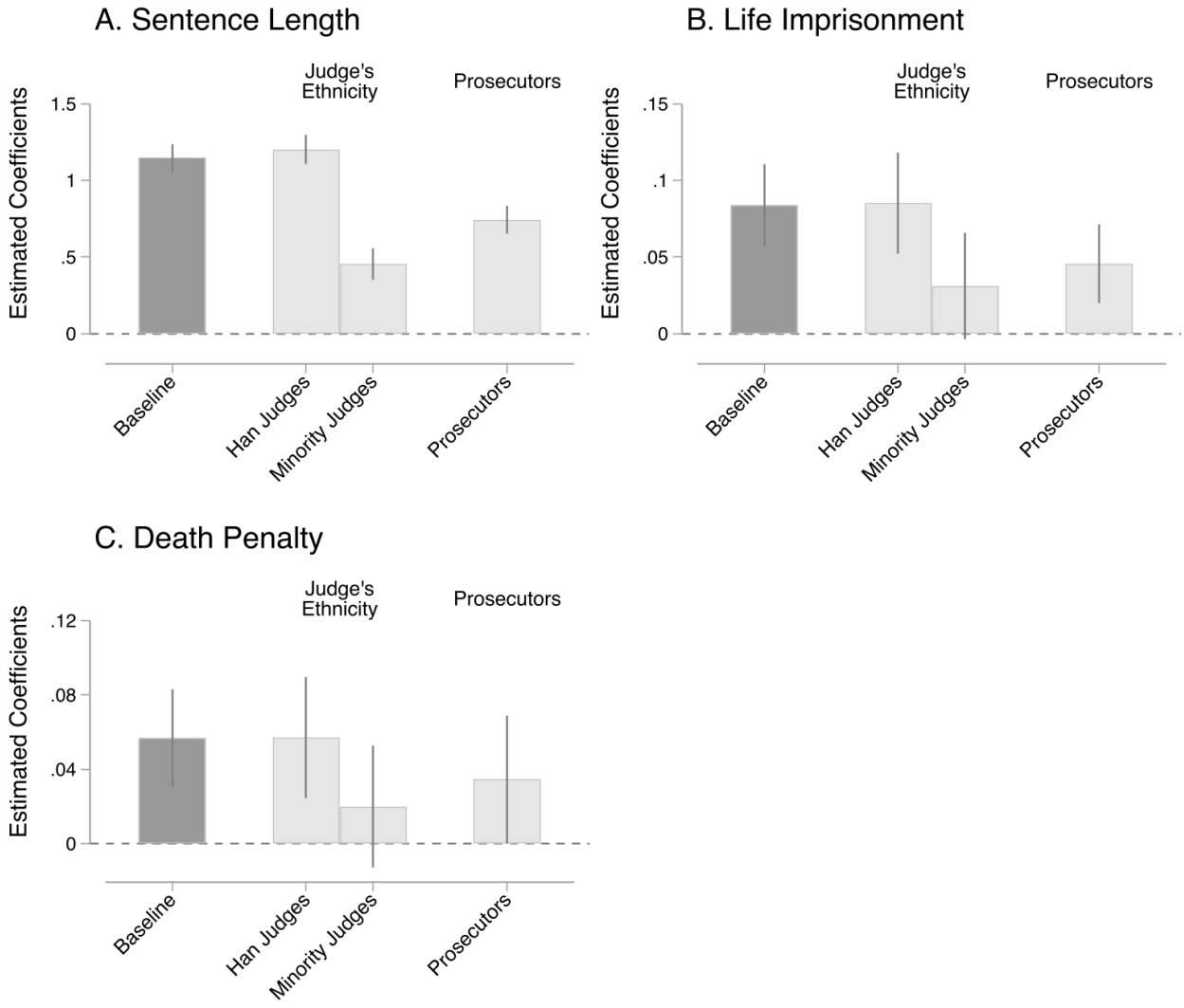
Notes: Eligible groups are the 20 largest ethnic minorities by national population in the 2010 census (NBSC, 2012, Table 1-13). Each panel displays the ten largest point estimates among these groups, without a significance cutoff. Each regression compares one minority group with Han defendants using equation (1), with the same controls and fixed effects. Panel A reports months; Panels B and C report percentage points and restrict the samples to eligible offenses. Numbers below the group names count minority defendants in each estimation sample. Bars show coefficients; whiskers show pointwise 95% confidence intervals with standard errors clustered by court.

Figure E2: Facts: Sentencing Disparity for Uyghur Ethnicity



Notes: This figure presents the additional penalties imposed on Uyghur and non-Uyghur ethnic minorities compared to Han defendants. For each group, we estimate equation (1) using only cases involving Uyghur or non-Uyghur minority defendants and the Han majority. The outcome variables are sentence length in months (Panel A), life imprisonment (Panel B), and the death penalty (Panel C). For life imprisonment and the death penalty, we restrict the samples to crime types eligible for these sentencing penalties. The bars represent the estimated coefficients, while the lines show the 95% confidence intervals. Standard errors are clustered at the court level. Panel D reports cases per population, plotting the number of criminal cases in our baseline data (Y-axis) against the population (X-axis) for each minority group.

Figure E3: Judge's Ethnicity, Prosecutors, and Minority Penalty



Notes: This figure presents key findings on the facts about ethnic minority issues in China's criminal justice system. The outcome variables are sentence length in months (Panel A), life imprisonment (Panel B), or the death penalty (Panel C). For life imprisonment and the death penalty, we limit the samples to crime types eligible for these sentencing penalties. The bars represent the estimated coefficients, while the lines show the 95% confidence intervals. Standard errors are clustered at the court level. For the judge's ethnicity, we estimate the baseline regression using each subsample. For prosecutors, we use the recommended sentences as the outcome variables. See the main text for additional details. Panels E and F of Table E1 report the estimates and standard errors.

Table E1: Facts about Ethnic Minority Issues in China's Criminal Justice System

	Sentence Length		Life Imprisonment		Death Penalty	
	Coef. (s.e.)	% of Han mean	Coef. (s.e.)	% of Han mean	Coef. (s.e.)	% of Han mean
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. All						
All Crimes	1.147***	6.8%	0.084***	6.5%	0.057***	14.0%
	(0.046)		(0.014)		(0.013)	
Panel B. Crime Types						
Violent Crime	1.462***	7.0%	0.101***	7.8%	0.072***	17.7%
	(0.068)		(0.025)		(0.022)	
Drug Crime	1.604***	5.3%	0.145***	11.2%	0.078***	19.2%
	(0.087)		(0.037)		(0.030)	
Sex Crime	1.406***	6.0%	0.084***	6.5%	0.038**	9.4%
	(0.096)		(0.023)		(0.019)	
Property Crime	1.113***	8.2%	0.036**	2.8%	0.024	5.8%
	(0.049)		(0.016)		(0.017)	
Others	0.342***	2.4%	0.022	1.7%	0.010	2.4%
	(0.050)		(0.020)		(0.015)	
Panel C. Ethnic Groups						
Christianity	1.552***	9.2%	0.111***	8.6%	0.075***	18.5%
	(0.056)		(0.030)		(0.024)	
Islam	1.255***	7.4%	0.080**	6.2%	0.067**	16.6%
	(0.067)		(0.036)		(0.027)	
Other Religions	1.000***	5.9%	0.032	2.5%	0.024	5.8%
	(0.047)		(0.024)		(0.021)	
Conflict History	1.482***	8.8%	0.120***	9.2%	0.101***	25.0%
	(0.057)		(0.029)		(0.021)	
Non-conflict History	0.939***	5.6%	0.047**	3.6%	0.028	6.9%
	(0.050)		(0.020)		(0.019)	

Panels D–F and notes continue on the next page.

Table E1 (continued)

	Sentence Length		Life Imprisonment		Death Penalty	
	Coef. (s.e.)	% of Han mean	Coef. (s.e.)	% of Han mean	Coef. (s.e.)	% of Han mean
	(1)	(2)	(3)	(4)	(5)	(6)
Panel D. Locations						
High Income	0.519***	3.2%	0.042**	3.2%	0.023	5.6%
	(0.054)		(0.018)		(0.018)	
Low Income	1.352***	7.7%	0.094***	7.3%	0.084***	20.8%
	(0.051)		(0.023)		(0.021)	
High Minority Share	1.406***	7.9%	0.088***	6.8%	0.077***	19.0%
	(0.048)		(0.015)		(0.016)	
Low Minority Share	0.598***	3.7%	0.046	3.5%	0.029	7.2%
	(0.050)		(0.033)		(0.020)	
High Crime Rate	1.307***	7.9%	0.090***	6.9%	0.068***	16.8%
	(0.047)		(0.019)		(0.017)	
Low Crime Rate	0.727***	4.2%	0.049**	3.8%	0.042**	10.4%
	(0.049)		(0.021)		(0.017)	
Panel E. Ethnicity of Judges						
Han Judge	1.201***	7.1%	0.085***	6.5%	0.057***	14.1%
	(0.048)		(0.017)		(0.017)	
Minority Judge	0.453***	2.8%	0.031*	2.4%	0.020	4.9%
	(0.052)		(0.018)		(0.017)	
Panel F. Prosecutor's Recommended Sentences						
Prosecutors	0.741***	3.0%	0.046***	3.5%	0.035**	8.5%
	(0.046)		(0.013)		(0.018)	

Notes: This table reports estimates of equation (1). Sentence length is measured in months; life-imprisonment and death-penalty coefficients and standard errors are in percentage points. Sentence-length percentages use the Han mean in each estimation sample. For life imprisonment and the death penalty, all rows use the Han rates in the respective all-crime baseline estimation samples: 1.300% and 0.405%. Court-clustered standard errors appear in parentheses. Life-imprisonment and death-penalty samples include only eligible offenses. Panel B estimates separate regressions by crime type. Panel C compares defendants from the minority groups in each religion or conflict-history category with all Han defendants. Panel D splits prefectures at the median income, minority population share, or per capita case count. Panel E splits the sample by judge ethnicity; Panel F uses prosecutors' recommended sentences as outcomes. All specifications include crime-category, year-month, province, court, and judge fixed effects and case and defendant controls. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

E-1 Baseline Estimates and Successive Controls

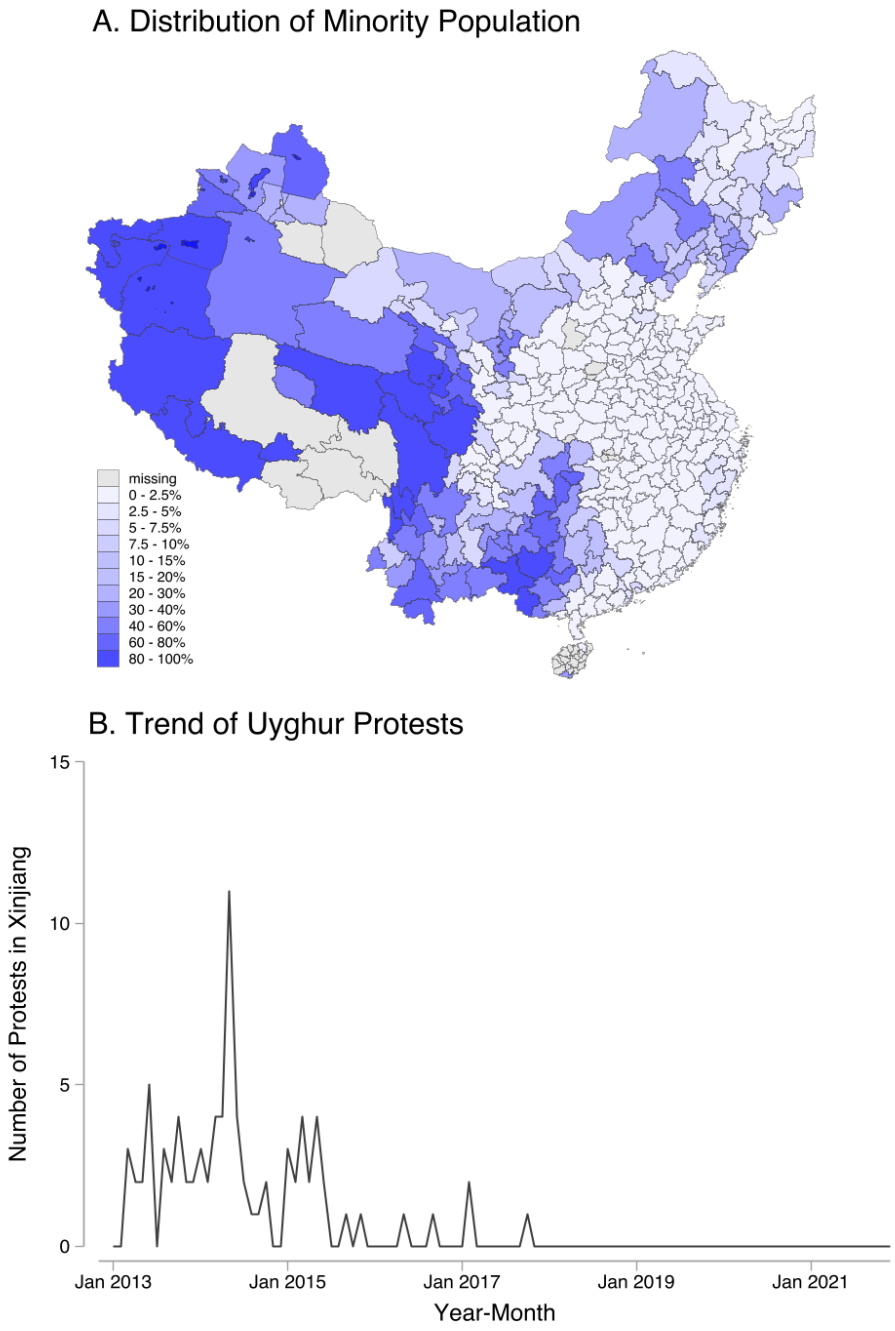
Table E2: Facts—What Contributes to Unconditional and Conditional Ethnic Disparity?

	Dependent variable: Sentence Outcomes					
	(1)	(2)	(3)	(4)	(5)	(6)
A. Effects on sentence length (months)						
minority	2.902***	0.711**	0.818***	0.676***	0.850***	1.147***
	(0.412)	(0.292)	(0.275)	(0.255)	(0.062)	(0.046)
% of Han mean	17.2%	4.2%	4.8%	4.0%	5.0%	6.8%
obs	6,440,961	6,439,124	6,439,124	6,439,124	6,439,118	6,434,145
B. Effects on life imprisonment (percentage points)						
minority	0.214	0.257**	0.242**	0.057	-0.062*	0.084***
	(0.139)	(0.126)	(0.113)	(0.088)	(0.034)	(0.014)
% of Han mean	16.4%	19.7%	18.6%	4.4%	-4.8%	6.5%
obs	1,451,469	1,450,699	1,450,699	1,450,699	1,450,680	1,446,093
C. Effects on death penalty (percentage points)						
minority	0.289***	0.190**	0.189**	0.108**	0.065***	0.057***
	(0.086)	(0.077)	(0.074)	(0.053)	(0.017)	(0.013)
% of Han mean	71.4%	47.0%	46.8%	26.6%	16.1%	14.0%
obs	1,963,066	1,962,011	1,962,011	1,962,011	1,961,995	1,957,249
Crime category FE	no	yes	yes	yes	yes	yes
Case characteristics	no	yes	yes	yes	yes	yes
Defendant characteristics	no	no	yes	yes	yes	yes
Year-month FE	no	no	yes	yes	yes	yes
Province FE	no	no	no	yes	yes	yes
Court FE	no	no	no	no	yes	yes
Judge FE	no	no	no	no	no	yes

Notes: This table reports the baseline results estimating equation (1). The outcome variables are sentence length in months (Panel A), life imprisonment (Panel B), and the death penalty (Panel C). For life imprisonment and the death penalty, we restrict the samples to eligible crime types and report coefficients and standard errors in percentage points. Relative gaps divide the coefficient by the Han outcome mean in the same estimation sample, using consistent units. We gradually add control variables and fixed effects from Column (1) to Column (6). Column (1) shows the unconditional gap, and Column (6) represents our baseline conditional gap presented in Table E1 and Figure 2. Standard errors are clustered at the court level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

F Supplementary Material for the Uyghur-Unrest Analysis

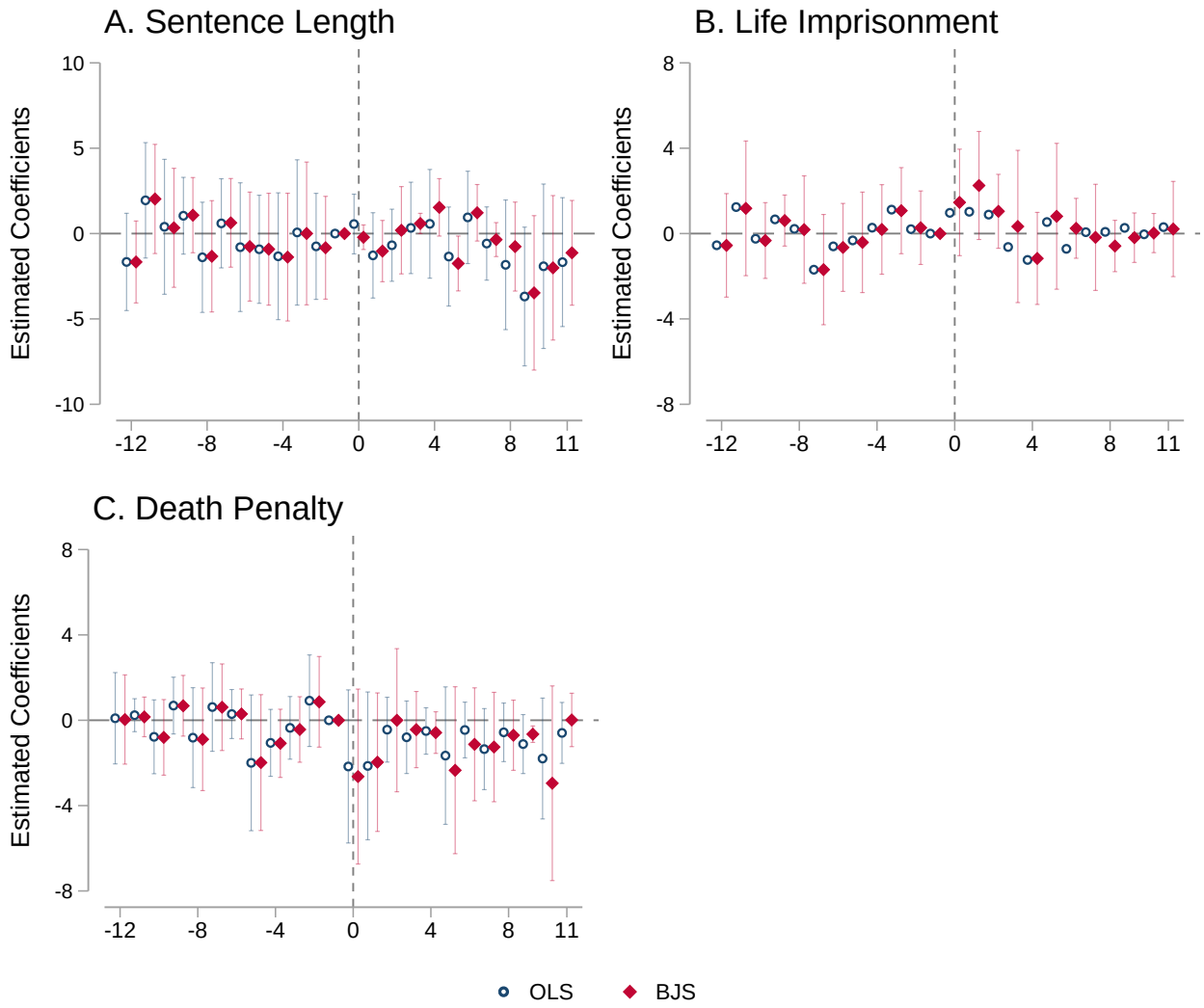
Figure F1: Variations in DDD: Minority Share and Uyghur Protests



Notes: This figure illustrates the variation used in the Uyghur unrest quasi-experimental design. Panel A shows the geographical distribution of minority population shares across prefectures. Panel B presents the time trend of Uyghur unrest in Xinjiang, China, from 2013 to 2021.

G Nationwide Sentencing Reform

Figure G1: Event Study of the Nationwide Sentencing Reform



Notes: The sample covers April 2016–March 2018; offense groups, outcome units, controls, and fixed effects follow Table G1. For OLS, we replace $Post_t$ in equation (G1) with indicators for each month and use March 2017 (month -1) as the reference; April 2017 is month 0. Each OLS coefficient measures how much more the minority–Han gap changed since March 2017 for targeted offenses than for comparison offenses. Whiskers show 95% pointwise confidence intervals with standard errors clustered two ways by province and primary offense. We show OLS life-imprisonment points without confidence intervals because the two-way covariance estimate is degenerate.

Table G1: Nationwide Sentencing Reform: Triple-Difference Estimates

	Sentence length (months) (1)	Life imprisonment (pp) (2)	Death penalty (pp) (3)
Panel A. OLS			
Targeted × Post × Minority	-0.549	0.073	-0.663
	(0.725)	(–)	(0.406)
Observations	1,554,335	493,392	603,906
Panel B. BJS			
Average imputed DDD effect	-0.598	0.355	-1.219
	(–)	(0.380)	(0.863)
Pretrend test p	0.897	0.788	0.662
Observations	1,556,583	492,764	603,827
Case characteristics	Y	Y	Y
Defendant characteristics	Y	Y	Y
Offense combination × Ethnicity FE	Y	Y	Y
Offense combination × Year-month FE	Y	Y	Y
Province × Ethnicity × Year-month FE	Y	Y	Y

Notes: The sample covers April 2016–March 2018; Panel A estimates

$$Sentence_{icpjt} = \beta Minor_i \times Targeted_i \times Post_t + \boldsymbol{X}'_{icpjt}\theta + \eta_{om} + \delta_{ot} + \gamma_{pmt} + \epsilon_{icpjt}. \tag{G1}$$

We follow the main text's notation, with $Targeted_i = 1$ for the original 15 covered offenses, $Post_t = 1$ from April 2017 onward, o denoting the offense combination, and m denoting minority or Han status. The terms η_{om}, δ_{ot}, and γ_{pmt} correspond to the three fixed-effect rows, and β measures how much more the minority–Han gap changed for targeted offenses than for comparison offenses. The comparison excludes the eight pilot-expansion offenses nationwide. Sentence length is measured in months; life imprisonment and death penalty are measured in percentage points (pp), using their respective eligible-offense samples. The control and fixed-effect rows apply to both panels. Standard errors, in parentheses, are clustered two ways by province and primary offense. Dashes indicate unavailable inference due to nonpositive or degenerate two-way covariance estimates. The BJS effect averages the post-reform months equally.